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Timestep-Conditioned Adapter Scaling for Multi-view Diffusion Texture Generation: A Controlled Empirical Study

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ABSTRACT

Geometry-conditioned multi-view diffusion provides an inference-time strength control, but its practical effect depends on temporal allocation, the conditioning interface, and the evaluated output. We conduct a controlled empirical study of Timestep-Conditioned Adapter Scaling (TCAS), carrying forward the historical middle-high profile C3 without further schedule selection. On 300 held-out objects with a frozen retrained GeoTex checkpoint, C3 improves foreground PSNR by 0.52 dB over fixed low, while competitive fixed and linear controls have better means on all four image endpoints. Matching applied-scale budgets changes the interpretation: C3 is 0.13 dB below fixed mean and 0.56 dB below late-high. Native MV-Adapter studies on 50 Fresh C inputs and 100 GSO objects detect endpoint-specific stage effects but do not detect the primary middle-window benefit over fixed mean. A ten-object native-UV bake pilot finds a small late-high LPIPS reduction of 0.00265 on unseen views; a separate twenty-object GSO evaluation with revised calibration does not detect this benefit. Generated-view and baked-output method rankings remain positively associated, although the preferred point estimates often differ. These findings quantify the effects and limits of geometry-control allocation and provide application-specific evidence for evaluating operating points. They support neither a universally preferred schedule nor physically faithful materials.

1. Introduction

Multi-view diffusion generates images that are projected onto a 3D mesh to form a texture. Geometry conditions help align the images with a known surface, while a reference image supplies appearance. The strength of geometric injection is an accessible inference-time control. Choosing it is an application decision: better generated-view agreement is useful only to the extent that the resulting textured asset retains appropriate appearance under the evaluation views.

The difficulty is that strength, temporal allocation and the final evaluation endpoint can change together. A schedule compared with a weaker constant may simply inject a larger scalar budget. A wrapper may cap some requested scales, leaving a different applied profile. A generated-view metric can also improve while color errors remain visible, and baking adds sampling and coverage losses. Establishing the value of a setting therefore requires competitive controls and actual textured-mesh evaluation.

We study these questions through Timestep-Conditioned Adapter Scaling (TCAS), an intervention on an existing GeoTex-Adapter residual interface. C3, the low–high–low profile recorded in the original development study, is retained without another selection search. Scheduling strength across time or layers is established prior work, including geometry-conditioning schedules in Scheduled Style Injection (SSI) [31]. Here the contribution is the quantitative evidence for geometry-guided multi-view texturing under stated interfaces and protocols.

The complete comparisons change the original practical recommendation. On 300 held-out objects, C3 improves over fixed low but loses to competitive fixed and linear controls. At equal applied-scale sums, fixed mean and late-high outperform its middle window. Native MV-Adapter evaluations on fifty Fresh C inputs and one hundred GSO objects do not detect the

primary middle-window advantage, although some secondary stage effects are measurable. Actual baking finds a small late-high benefit on ten compatible GeoTex objects and no detected LPIPS benefit on a separate twenty-object GSO cohort. These findings delimit when an allocation result has application value.

The contributions are:

(1) **A controlled assessment of strength and temporal allocation.** Matched comparisons with competitive fixed and linear controls distinguish the recorded C3 intervention from a timing-only interpretation. Equal per-group applied sums identify stage-allocation total effects within the frozen GeoTex pipeline.

(2) **An application-level evaluation of those effects.** Actual native-UV meshes, matched-resolution GT self-bakes and eleven unseen cameras quantify small local gains, persistent appearance errors and changes in the preferred operating point. Both the ten-object gain and the twenty-object non-detection are reported.

(3) **An evaluation of interface and protocol boundaries.** Native MV-Adapter matrices and retained MVDiffusion results test related allocation questions on different conditioning operators. They reveal endpoint-dependent effects and limits to transfer, without pooling cohorts or claiming a common residual dose.

Object-level records and runnable analyses support these findings. TCAS remains the study intervention and C3 a historical operating point; neither the scheduling concept nor the existence of an audit archive is presented as an independently new algorithm.

2. Related Work

2.1. 3D Texture Generation and Multi-view Diffusion

Single-image 3D generation is often decomposed into geometry generation and texture generation. Single-image-to-3D, multi-view diffusion, and sparse-view reconstruction methods have improved structural stability and view synthesis [1, 2, 3, 4, 5, 6, 7, 8]. Texture synthesis remains a distinct challenge for asset realism and usability [15]. Many systems generate RGB views with 2D diffusion and project or bake them onto a surface [9, 10, 11, 12]. Recent work also explores exemplar-based texture synthesis and diffusion-based procedural materials [16, 17].

Paint3D, MVPaint, Hunyuan3D 2.0, and MVPainter use geometry-conditioned multi-view generation in 3D texturing pipelines [18, 19, 20, 13]. MV-Adapter, MVDiffusion, and other synchronized multi-view methods address consistency through different adapter and denoising designs [21, 22, 23, 24]. These systems establish the graphics context for this work. The present evidence does not compare final baked-texture quality across those systems.

2.2. Adapter-Augmented Diffusion Models and Geometric Control

ControlNet and T2I-Adapter provide learned spatial-conditioning branches or feature injections, whose contribution can be weighted by an inference-time condition-strength scale [25, 26]. They establish conditioning strength as a practical control, but their injected tensors and scaling sites do not share a numerical dose with GeoTex. The studied GeoTex-Adapter path already exposes an inference-time residual scale and evaluates fixed requests. TCAS reuses that control point as a three-stage request over sampling progress, while recording how wrapper-group caps change the applied values and how adapter activations determine the resulting corrections. It adds no learned branch or schedule selector. IP-Adapter introduces image prompts through decoupled attention [27]; LoRA and related methods use scaled low-rank updates [28, 29, 30].

2.3. Diffusion Guidance Scheduling and Timestep Control

Diffusion denoising is often discussed in terms of coarse-to-fine evolution, and prior work varies guidance strength over sampling progress to study quality, diversity, or fidelity [32, 33, 34, 35]. Limited-interval classifier-free guidance is one example; it changes the combination of conditional and unconditional score predictions [14].

SSI is the closest scheduling precedent. It varies StyleID's query-mixing parameter across decoder layers and denoising time, tests schedule direction and shape, and applies scheduling to a separate geometry-conditioned ControlNet branch; its reported ControlNet scheduling gains over fixed conditioning are modest relative to its style-injection results [31]. This prior work already covers the general idea of layer/time scheduling and geometry-conditioning schedules. TCAS acts on the residual correction produced by GeoTex-Adapter from normal, depth, and foreground-mask conditions, with group-specific wrapper caps; it does not modulate StyleID attention mixing or add a ControlNet branch. The task, interface, and measured endpoints differ, so SSI does not imply the Fresh C paired estimates, the requested/applied/residual accounting, or the observed image- and color-agreement failures. These task-specific measurements are the empirical contribution; the scheduling concept itself is not claimed as new.

2.4. Multi-view Adapter Interfaces

MV-Adapter uses a condition guider for camera or geometry maps together with decoupled spatial, multi-view, and optional image cross-attention layers [21]. MVDiffusion uses correspondence-aware attention modules to connect pixels across views [22]. These interfaces differ from the GeoTex-Adapter residual branch studied here. We therefore report each auxiliary intervention within its own tested implementation and cohort, without pooling metrics or interpreting it as an architecture-matched replication of TCAS. The native temporal comparisons are summarized in Tables 6 and 7; historical panels remain in Supplementary Section S3.

3. Study Design and Control Interface

3.1. Task Definition and Base Pipeline

This paper builds on the MVPainter-style geometry-controlled multi-view texture generation framework [13] and studies how the inference-time strength of geometric condition injection affects the generated results. Figure 1 shows the

TCAS control gate on the GeoTex-Adapter residual path: the requested scale is limited by the wrapper's group-specific cap before the residual is added to frozen UNet features. Given a single reference image and the corresponding 3D geometry, the base pipeline generates RGB texture images from six target views and can pass them to downstream projection or texture baking. Fresh C evaluates generated target views; the ten-object GeoTex pilot and twenty-object GSO evaluation separately assess unlit baked meshes. We do not modify the base UNet, VAE, or pretrained adapter weights.

During multi-view diffusion, the 3D geometry is rendered into normal maps, depth maps, and foreground masks, denoted as:

$$\mathbf{G} = \{G_{\text{normal}}, G_{\text{depth}}, G_{\text{mask}}\}. \tag{1}$$

The geometric adapter maps these conditions to a residual correction term and injects it into the intermediate hidden states of the UNet. For inference step j with normalized execution progress p_j, the adapter injection can be written as:

$$\mathbf{h}'_{j,\ell} = \mathbf{h}_{j,\ell} + s_{\text{app}}(p_j, \ell)\, A^{\ell}_{\phi}(\mathbf{h}_{j,\ell}, \mathbf{G}), \tag{2}$$

Here ℓ indexes an injected layer group. The *requested scale* $s_{\text{req}}(p_j, \ell)$ is the value assigned by the profile; the *applied scale* is the value used by the wrapper,

$$s_{\text{app}}(p_j, \ell) = \min(s_{\text{req}}(p_j, \ell), c_\ell), \tag{3}$$

when the native cap c_ℓ is active, and otherwise equals the request. The resulting correction is $\mathbf{r}_{j,\ell} = s_{\text{app}}(p_j, \ell) A^{\ell}_{\phi}(\mathbf{h}_{j,\ell}, \mathbf{G})$; its *post-scale residual magnitude* is $m_{j,\ell} = \|\mathbf{r}_{j,\ell}\|_2$. Thus neither a requested nor an applied scale label alone determines the correction magnitude.

In the audited Fresh C wrapper, the three cap groups map the GeoTex residual wrappers on the UNet ResNet blocks as follows: deep comprises the UNet middle block and `up_0`, middle is `up_1`, and shallow is `up_2`. These names identify the implementation's grouped injection points; they do not assert universal semantic roles for denoising stages. The corresponding Fresh C caps are 3.0, 3.5, and 0.8. C3's shallow request of 1.25 is therefore applied as 0.8, while its deep and middle requests pass unchanged. GFL and GFH also clip at the shallow cap. These cap values describe the audited Fresh C wrapper; row-level cap semantics for the historical 24-object probe are not fully recoverable. TCAS scales a geometry residual added to feature activations; it does not reweight conditional and unconditional score predictions as classifier-free guidance does.

3.2. Image-Space Metrics and Texture-Variation Diagnostics

The endpoints describe different outcomes. *Image-space agreement* refers to target-view PSNR, LPIPS, and SSIM measurements on the named image regions; it is not a measure of final 3D material quality. *Reference-color agreement* refers to color differences such as CIEDE2000 computed against the corresponding reference. *Texture variation* refers to statistics such as RGB standard deviation, Laplacian variance, gradient magnitude, and high-frequency energy; these may rise with detail or artifacts and are not fidelity measures. *Subjective preference* refers only to the submitted forced-choice study. *Baked 3D appearance* refers to rendered mesh outputs after texture projection or baking, while *PBR/material fidelity* additionally concerns appearance under the material and illumination model. These endpoints are not interchangeable. FG-SSIM, Edge-SSIM, and Crop-SSIM quantify structural similarity on the indicated image regions; PSNR measures pixel error; LPIPS compares learned image features [36, 37].

Laplacian variance, RGB standard deviation, and gradient magnitude can describe local high-frequency or color variation. For example, a foreground Laplacian statistic may be written as

$$M_{\text{var}} = \text{Var}(\text{Lap}(I_{\text{out}} \odot G_{\text{mask}})), \tag{4}$$

where G_{mask} is the foreground mask. A larger value is not necessarily better: detail, noise, repeated patterns, or artifacts can all increase local variation. We therefore do not interpret these proxies as texture fidelity or use them alone to claim that a schedule better preserves reference appearance.

3.3. TCAS Control Semantics and Comparison Design

TCAS replaces a globally fixed requested adapter scale with a piecewise profile over normalized sampling progress. The early/middle/late partition is an operational description, not a universal claim about the function of every denoising stage. C3's low–high–low profile is the specific development choice evaluated in this revision; it is not presented as an independently derived optimum.

To avoid ambiguity caused by different raw timestep directions in different schedulers, $p_j = j/(T-1)$ denotes execution progress for inference step $j \in \{0, \ldots, T-1\}$. Thus $p = 0$ is the first update from the initial noisy latent and $p = 1$ is the last denoising update before decoding. For $T = 50$, TCAS divides execution into steps 0–16, 17–32, and 33–49 (17/16/17 steps):

$$s_{\text{req}}(p) = \begin{cases} s_e, & 0 \le p < 1/3 \\ s_m, & 1/3 \le p < 2/3 \\ s_l, & 2/3 \le p \le 1 \end{cases}. \tag{5}$$

We call this variation across the 50 inference steps the *temporal allocation*. A *layer allocation* varies the requested scale across the three wrapper groups at a fixed step. For a schedule with applied values $s_{\text{app}}(j, \ell)$, its per-group applied-scale sum is $B_\ell = \sum_j s_{\text{app}}(j, \ell)$; the unweighted sum $B_{\text{nom}} = \sum_\ell B_\ell$ is a *nominal applied-scale budget*. It does not weight scales by adapter activations. We log post-scale correction summaries, including wrapper-level L_2 norm and mean absolute entry. The three wrapper groups operate on feature maps with different spatial dimensions, and the traces do not retain channel counts; raw L_2 values are therefore not used for cross-layer dose comparisons. The per-entry mean absolute correction reported in Supplementary Section S1 is a dimension-normalized descriptive summary, not an equal-dose intervention. Correction magnitudes also depend on the current hidden state and earlier denoising updates, so equal nominal budgets do not imply equal realized corrections.

The selected C3 configuration used in this paper is:

$$(s_e, s_m, s_l) = (1.25, 2.50, 1.25). \tag{6}$$

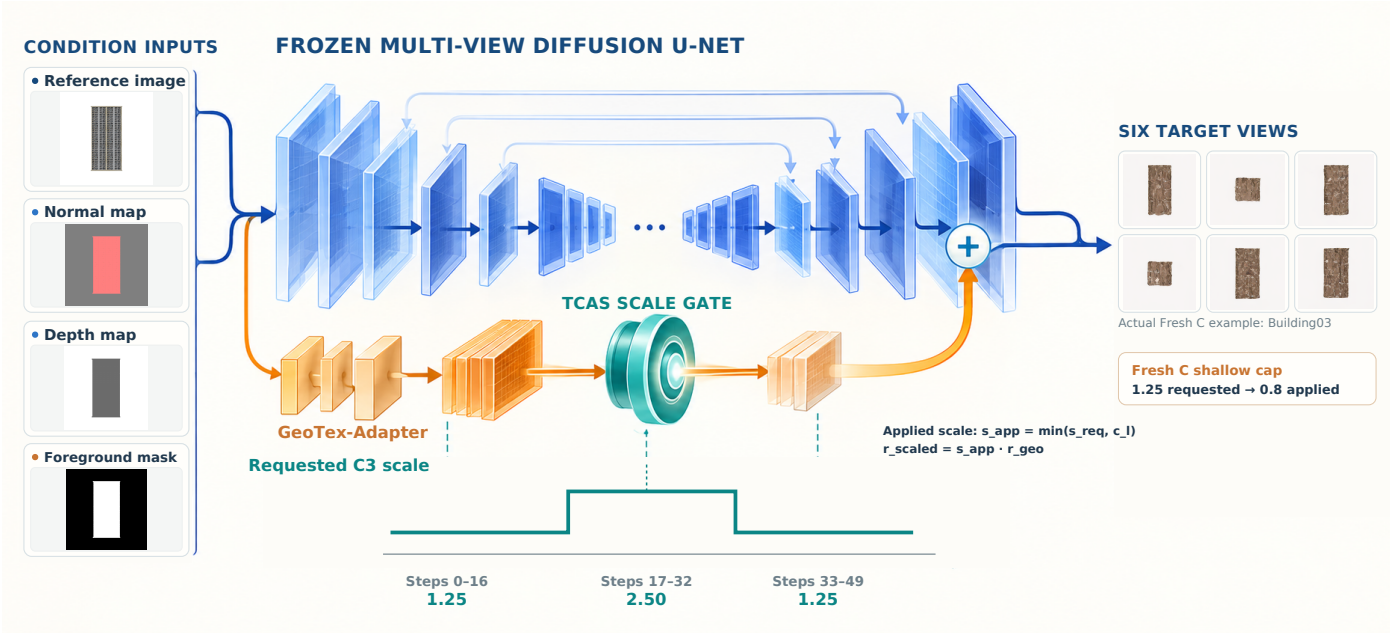


Fig. 1. TCAS on the GeoTex-Adapter residual path. The requested C3 profile is (1.25, 2.50, 1.25) **over steps 0–16, 17–32, and 33–49; the wrapper applies the request after group-specific native caps, then adds the scaled residual to frozen UNet features. In Fresh C, the shallow-group cap is 0.8. Input maps and six C3 output thumbnails are actual Fresh C Building03 data (Shalmon, CC BY 4.0; Supplementary Section S9), used here to illustrate the interface. This control schematic is not an end-to-end application result. Conceptual artwork was generated with OpenAI Images (version not recorded by the tool) and assembled with reproducible vector labels; it does not encode measured results. Baking is evaluated separately in Section 4.5.**

The low–high–low profile is a concrete inference-time intervention motivated by the submitted development analysis. Its stage interpretation is a task-specific hypothesis: the tested early, middle, and late allocations may produce different responses. Fresh C evaluates the recorded profile against matched controls but does not prove a universal function for any stage. TCAS adds no trained parameters and does not modify base-model or adapter weights; it replaces the requested scale during inference.

What C3 changes and what its comparisons identify. C3 requests (1.25, 2.50, 1.25) for each of the 50 steps across the three layer groups before native caps. In Fresh C, the applied deep and middle profiles retain those values; the shallow profile is clipped to 0.8 throughout. GFL requests 1.25 at every step and group, so its deep and middle values are 1.25 throughout and its shallow value is also clipped to 0.8. Their per-group applied-scale sums are therefore 82.5 versus 62.5 in deep and middle, and 40 for both in shallow. The C3–GFL contrast changes the middle-third deep/middle scale magnitude as well as its temporal allocation; it is not a timing-only comparison.

To identify a temporal-placement contrast, controls must hold the per-layer applied values and nominal per-group sums fixed while moving the same values between time windows. To identify a scale-strength contrast, controls must hold the layer and temporal allocations fixed while changing amplitude. Both contrasts also require reporting post-scale correction summaries: matching requested values or applied-scale sums does not match the evolving residual corrections. Raw L_2 values are dimension-sensitive across wrapper groups and cannot establish cross-layer dose matching. The current 300-object allocation comparison exactly matches the applied sums in each layer group. This controls the scalar budget for the total allocation effect, although it does not identify a residual-dose-independent mechanism. Post-hoc adjustment for state-dependent corrections cannot by itself recover that mechanism. The older strict-276 analysis is reported separately.

Development origin and scope of C3. The submitted 01549 paper records C3 = (1.25, 2.50, 1.25) as the selected profile from a 24-object development probe containing 17 uniform, layer-wise, and time-conditioned variants. We retain that development record because it explains the origin of the profile studied here. The complete row-level outputs and selection-time provenance are not available in the current evidence authority, so the historical ranking and labels cannot be independently checked. We therefore make no formal optimality or uniqueness claim based on this probe. Fresh C evaluates the recorded profile without further schedule search.

3.4. Geometry Adapter and Inference Interface

GeoTex-Adapter is treated as the fixed geometry-residual backbone, not as a new contribution of this paper. It receives a five-channel input comprising a three-channel normal map, a one-channel depth map, and a one-channel foreground mask. A multi-scale encoder produces residual features that are injected into the frozen UNet. TCAS acts on the requested multiplier along denoising progress; it does not retrain the adapter, the UNet, or the VAE and introduces no learned parameters.

Previously reported Full Adaptive Correction (FAC) training is retained as an auxiliary negative result in Section 4.7. It is distinct from the frozen inference interventions evaluated here.

The recorded analysis and display code uses fixed object-level inputs and the original comparison families. Actual experimental images are preserved without appearance retouching. Software identities and the AI-assistance disclosure are supplied with the reproduction materials.

4. Experiments

4.1. Study Cohorts, Sampling and Statistical Units

Table 1 separates the studies and their evidence classes. Fresh C contains 300 UIDs disjoint from the historical schedule-selection/evaluation pool, the recorded 1,118-object training list and the earlier clean-v2 evaluation list. Its primary four-condition comparison uses one frozen controlled-retraining checkpoint, shared inputs and per-object seeds. It does not reproduce the historical adapter's reported +0.96 dB estimate. The original checkpoint's exact identity remains unavailable.

Within each comparison, object inputs, targets, initialization and endpoint code are shared. Fresh C uses seed 42 + `object_idx`, held constant across conditions for that object. Its capped residual groups have deep/middle/shallow limits of 3.0/3.5/0.8. MV-Adapter uses a different feature injection operator and numerical scale range; its native target cameras and actual default-path identity are checked separately. Scale labels are not common physical doses across interfaces.

Fresh C's controlled sampler differs from the complete base pipeline, which uses online image features, classifier-free guidance and 512-pixel views. Constant 1.0 under the GeoTex caps also differs from the unset-scale residual branch. We therefore name these controls by their executed configuration. The source records and sampling/default-path diagnostics are in Supplementary Sections S1 and S6.

FG-PSNR, LPIPS, SSIM and Edge-SSIM measure agreement on the named image regions. Baked evaluation renders actual exported meshes; unlit base color does not establish PBR or material fidelity. Views are averaged within object, with GSO sampling seeds also averaged within object. Paired intervals resample objects and are pointwise 95% intervals. Preserved primary, secondary and exploratory families use their stated Holm adjustments; the distinct families do not imply one overall experiment-wide error guarantee. Non-detection is not equivalence or non-inferiority. Complete comparison rules are supplied in Supplementary Sections S2–S5.

4.2. Development origin and historical scope

The historical study selected C3 from 17 profiles on a 24-object probe and reported a separate fixed-scale sweep. Those records explain the intervention's origin, but their complete row-level provenance cannot be reconstructed. They remain historical diagnostics in Supplementary Section S7. The current matched evaluation carries C3 forward without another selection search. Historical pooled gains, preference and FAC results remain separate from the new evidence.

4.3. Competitive Controls and Matched Temporal Allocation

The internally specified Fresh C primary is C3 minus Fixed Low (GFL) FG-PSNR. The paired gain is +0.52 dB (95% interval [+0.40, +0.64]), with 206/300 objects favorable. All conditions use the same frozen retrained checkpoint and object inputs. This gain combines increased strength and temporal allocation: C3's deep/middle applied sums are 82.5 each, versus 62.5 for GFL, while both have shallow sum 40.

Competitive controls change the recommendation. Fixed High has better means than C3 on all four endpoints: C3 minus Fixed High is −0.748 dB FG-PSNR (pointwise interval [−0.885, −0.609]), with FG-SSIM difference −0.04553 ([−0.04937, −0.04167]). Generic Linear ramps deep/middle requests from 1.25 to 2.50 and shallow requests from 0.50 to 0.75 over the fifty steps. Its applied sums are (93.75,93.75,31.25), so it changes layer allocation and total budget as well as timing. It also has better means on all four; C3 minus Generic Linear is −0.593 dB ([−0.837, −0.351]). Its positive median of +0.074 dB and 155/300 favorable objects show why mean and object-level distributions must both be considered. Complete original paired endpoints and their comparison families are in Supplementary Section S2.

Equal applied budgets.. Early-, middle-, and late-high move the same sixteen high steps to updates 0–15, 17–32, and 34–49. Deep/middle low and high are 1.25 and 2.50; shallow stays at its 0.8 cap. Fixed Mean uses 1.65 before the cap. Each allocation therefore has per-group sums $(82.5, 82.5, 40)$. The completed six-condition matrix contains 1,800 paired rows with common-input and seed checks.

C3 minus Fixed Mean is −0.133 dB (pointwise interval $[-0.170, -0.096]$); C3 minus Late-high is −0.564 dB ($[-0.634, -0.494]$). C3 also trails both on LPIPS, FG-SSIM and Edge-SSIM. It improves over Early-high by approximately 0.50 dB. Within these six conditions Fixed High wins FG-PSNR on 172 objects, Late-high on 64, and C3 on none; this is an endpoint-specific count, not a universal quality ranking.

The equal-budget comparison identifies an allocation total effect. Evolving residual amplitudes remain trajectory-dependent, preventing a residual-dose-independent mechanism interpretation. Late-high is not universally preferred: Fixed Mean has higher FG-SSIM, and Generic Linear is a strong unmatched-budget practical control. The historical LLH layer/time addendum and complete object distributions remain in Supplementary Section S2.

Table 1. Studies, implemented protocols and evidence scope. Rows are studies rather than disjoint populations: the bake pilots and some interface tests reuse image-study inputs. Object is the statistical unit; views and seeds are nested within object.

Study / objects	Interface and sampling	Endpoint	Selection and scope
Fresh C / 300	Capped GeoTex residual; 256 pixels; 50 Euler steps; conditional-only prediction; cached image features	Six target-view foreground endpoints	UID-disjoint held-out cohort; original primary C3–GFL frozen; linear and equal-budget comparisons are same-cohort addenda
GeoTex bake / 10	Same controlled generator; six 256-pixel bake cameras; native UVs	Eleven unseen unlit views	Input/compatibility-selected subset; primary Late–Mean specified before new bake outcomes; two later native controls exploratory
MV-Adapter / 50	Native condition-guider features; 512 pixels; 50 steps; CFG 3	Six native-camera target views	Fresh C inputs not previously viewed through this interface; input-only selection; thirteen legal white targets retained
GSO / 100	Native MV-Adapter; 512 pixels; 50 steps; CFG 3; three seeds	All-six PSNR primary; five nonreference LPIPS secondary	Source-ordered convenience cohort; four inherited profiles frozen after earlier baseline exposure; external sensitivity
GSO bake / 20	Same fixed GSO inputs; native UVs; three seeds	Eleven unseen unlit views	Revised GT-only calibration after fifteen original outcomes; full twenty retained; no GT-error subtraction
Reference-protocol / 50	New frozen GeoTex checkpoint; native reference conditions; 512 pixels; 50 steps; CFG 2	Five nonreference views	Supporting protocol/checkpoint sensitivity on previously studied inputs; not another independent backbone
MVDiffusion / 75	Partial correspondence-block control; depth concatenation unscaled	Recorded six-view endpoints	Retrospective third-interface panel; narrow native-mirror identity check; not exact GeoTex operator transfer

Table 2. Fresh C competitive-control means on the same 300 objects. Early/Middle/Late-high and Fixed Mean share exactly the same per-group applied sums. The equal-budget study is a post-inspection addendum, not a second holdout. Each budget entry for deep/middle applies separately to both groups. †Generic Linear is a competitive same-cohort addendum, not part of the matched-budget family; its different group budgets are shown explicitly.

Control	Deep/middle sum	Shallow sum	FG-PSNR↑	FG-LPIPS↓	FG-SSIM↑	Edge-SSIM↑
Fixed Low	62.5	40	11.31	0.2051	0.3500	0.4728
Fixed High	125.0	40	12.58	0.1891	0.4201	0.4961
Fixed Mean	82.5	40	11.96	0.1978	0.3864	0.4843
Early-high	82.5	40	11.33	0.2045	0.3696	0.4765
C3 / Middle-high	82.5	40	11.83	0.1998	0.3746	0.4773
Late-high	82.5	40	12.39	0.1951	0.3788	0.4914
Generic Linear†	93.75	31.25	12.42	0.1862	0.4026	0.4880

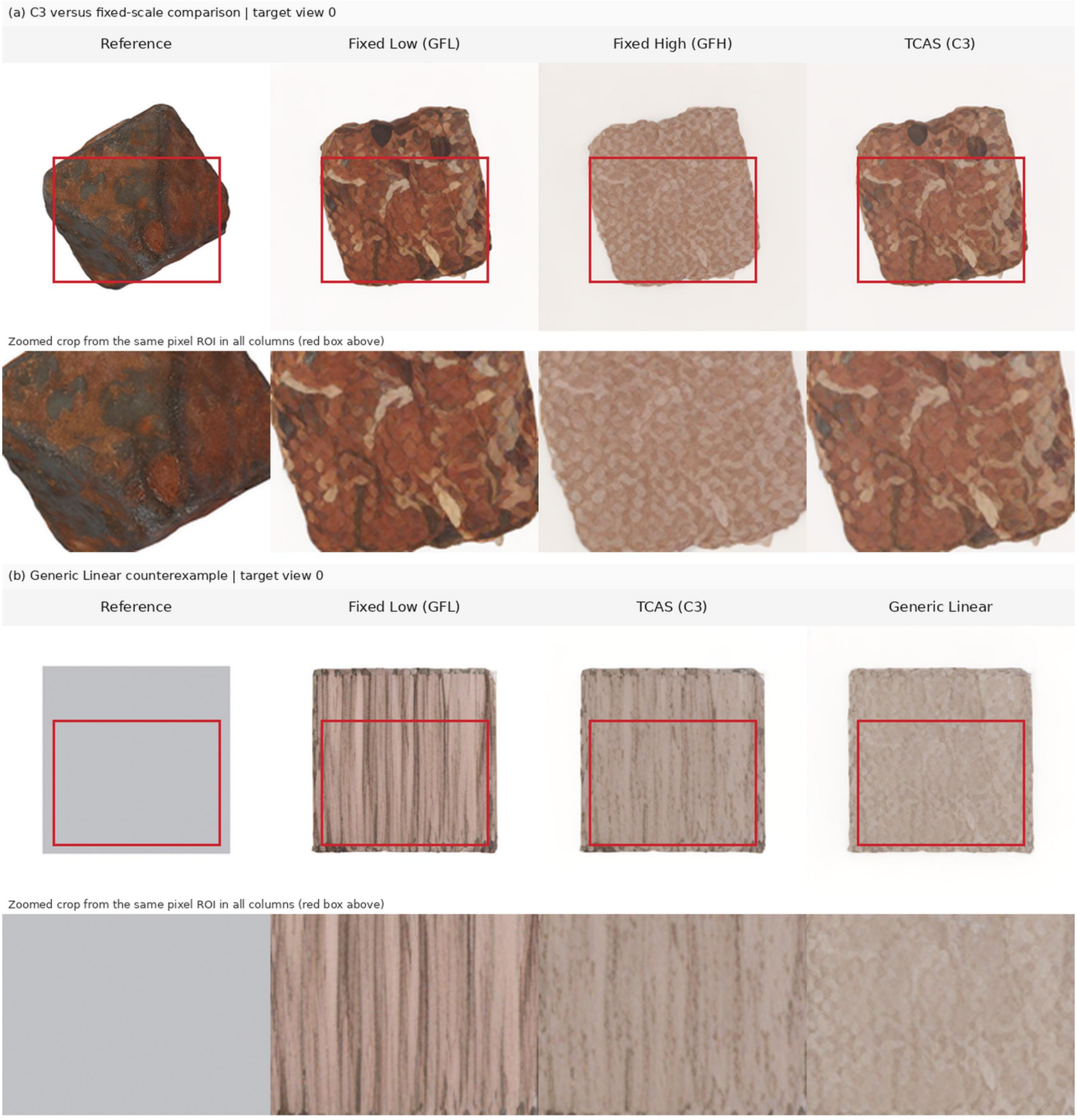


Fig. 2. Matched Fresh C target-view RGB outputs for the same frozen object, reference, camera, checkpoint, runner, seed policy, and inference protocol. (a) shows a selected fixed-scale case with an endpoint trade-off between C3 and GFH. (b) shows a counterexample where C3 minus Generic Linear is -3.491 **dB FG-PSNR and** $+0.08172$ **FG-LPIPS, favoring Generic Linear on both endpoints. Each context marks the exact pixel ROI shown below. Outcome-selected examples do not estimate prevalence or texture fidelity; the RGB crops do not establish colorimetric agreement. Full six-view panels remain in the retained RGB displays in Supplementary Section S9; model credits and image/ROI provenance are in Supplementary Section S9 and the Fresh C figure manifest.**

4.4. Appearance and Color Failure Analysis

The selected color diagnosis contains 26 failure-enriched objects: six earlier figure cases and 20 Fresh C panels. It is not a random sample, a new holdout, or a population estimate. No Adapter, GFL, and LLH were compared on the same inputs, checkpoint, and runner with six target views per object (78 object-condition records). The following paired object-bootstrap intervals are descriptive (20,000 resamples; seed 61008), and no confirmatory multiplicity family was defined.

GFL lowered reference-relative CIEDE2000 in 24/26 objects, while visible pink or purple shifts remained in two legacy Fig. 4 cases under No Adapter, GFL, and LLH. Their mapping to the original submitted panels cannot be recovered; the traceable Fresh C example in Figure 3 is a different object and is not presented as a reproduction of those cases. The decoded RGB output showed the shifts before UV projection or baking, and a late denoising snapshot (update 40 of 50) showed one of them. In that case, a ground-truth VAE encode–decode isolate had CIEDE2000 4.086 and $\Delta a^* \approx -1.35$, compared with 22.032 and +1.977 for GFL. This weakens ordinary VAE round-trip error as the sole explanation in that case; it does not identify the complete cause or rule out other model and conditioning effects.

LLH–GFL increased mean CIEDE2000 while reducing foreground gradient magnitude, high-frequency energy and color entropy. Lower variation therefore did not imply closer reference appearance. The denoising/runtime isolates and three failed correction candidates remain in Supplementary Section S6; they localize observations but do not validate a cause or repair.

Table 3. Selected paired results from the failure-enriched 26-object color diagnosis. Negative CIEDE2000 differences indicate lower reference-color error for the first condition. The set is descriptive and does not estimate population prevalence.

Contrast	Mean CIEDE2000 difference [95% CI]	Objects
GFL – No Adapter	−19.654 [−26.370, −13.248]	24/26 lower
LLH – GFL	+3.583 [+0.462, +6.712]	–

Historical twenty-object results, the earlier five-object bake, invalidated camera/mask attempts and the corrected REPEAT sampler are reported separately in Supplementary Section S5. They are not pooled with the new ten-object cohort. The actual unset-scale GeoTex residual branch differs from Constant 1.0 (capped), so the latter is not labeled an unmodified pipeline.

Full-object evidence and its selection.. The strong-fixed and Generic Linear examples in Figure 2 and the color failure in Figure 3 use common framing and exact shared pixel crops. Favorable/adverse C3–GFL examples are retained in Supplementary Section S9. Outcome-selected displays illustrate behavior, not prevalence. The unrecoverable historical texture audit's 92% prevalence claim and its texture-loss classifications are withdrawn; variation statistics alone would not validate fidelity even if their lineage were complete.

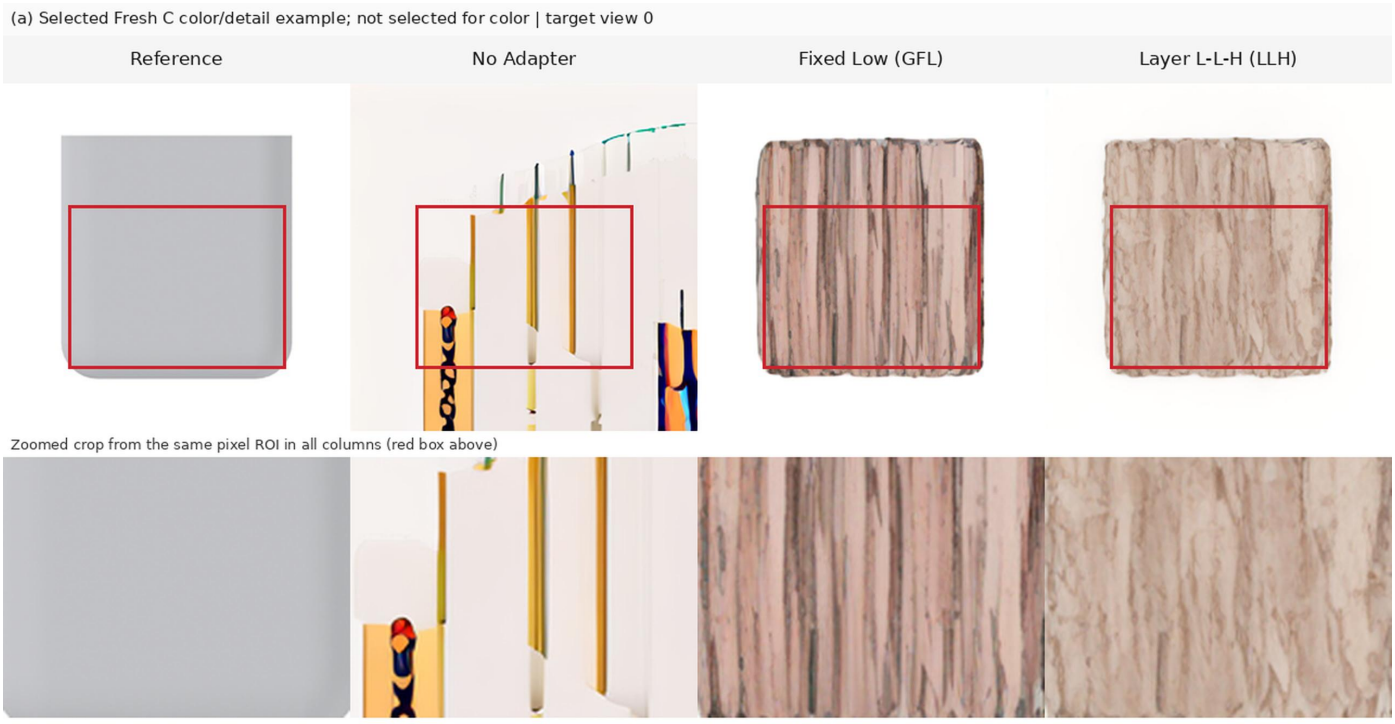


Fig. 3. Matched Fresh C target-view RGB outputs for UID `f6979d7ff8e54c4daf00fee46f09a139`: reference, No Adapter, GFL, and LLH. Each column shows the same target view (view 0), full context, and shared pixel ROI. This previously viewed object is one of the 20 Fresh C panels in the 26-object diagnosis; its frozen stratum was the lowest C3–GFL FG-LPIPS, not color-selected. It is not independent confirmation or a population estimate, and it is distinct from the unrecoverable legacy 01549 color panels. Full attribution and source/ROI hashes are listed in Supplementary Section S9 and the figure manifest.

4.5. Actual Baked Appearance: Local Benefit and External Limits

We froze ten previously unbaked Fresh C objects in UID order using input availability, native UV/material compatibility, triangle-count and camera-mask alignment criteria. Of the 300 screened inputs, only ten passed all criteria before the cutoff; the planned maximum was twenty. The ten are not new to the GeoTex image domain, but their baked outcomes had not been inspected during selection. This is a compatibility-selected pilot, not a representative twenty-object confirmation. Five earlier bake objects are reported separately.

Seven conditions share each object's source mesh, reference, seed, six projection cameras, masks, 256-pixel source resolution, atlas resolution, hole filling and eleven unseen evaluation cameras. Original geometry, UVs and legal base-color material semantics are retained and checked. Unseen cameras do not enter the bake. GT-256 uses the same source resolution, masks and bake parameters and has exactly the same raw atlas coverage as every generated condition. The separately tested CPU REPEAT sampler passes boundary and integer-period tests; it is not the sampler used by this GPU bake. Pole-view RGB orientation is corrected without rotating raw-camera masks. These calibrations control projection artifacts; evaluation is unlit base color, not PBR fidelity. Supplementary Section S5 records the checks and object-level data.

The internally predeclared primary contrast was Late-high minus Fixed Mean on unseen-view FG-LPIPS, proposed from the earlier pilot and tested on the ten newly selected bake objects. Its mean is −0.00265 (95% object-bootstrap CI [−0.00414, −0.00137]; exact paired sign-flip $p = 0.00195$), with all ten objects favorable. This is approximately a 2.0% reduction relative to Fixed Mean's mean LPIPS, and identifies a small allocation benefit within this sampling protocol. Generic Linear has a slightly lower mean LPIPS but lower PSNR; Late-high does not dominate every endpoint or control. The matched GT-256 reference bake has mean LPIPS 0.0590, versus 0.1303 for Fixed Mean and 0.1277 for Late-high. A post hoc description therefore puts the absolute gain in context: it closes only 3.7% of the mean Fixed Mean–GT reference gap (pointwise

Table 4. New ten-object unlit unseen-view bake pilot. Eleven views are averaged per object. Higher PSNR/SSIM and lower LPIPS/CIEDE2000 are favorable. Constant 1.0 is an explicitly capped request, not the unset-scale pipeline. All seven original generated conditions and all ten eligible objects are retained; the two subsequently added controls marked † are exploratory; GT-256 is a separate matched reference bake, not a theoretical bound.

Condition	FG-PSNR	FG-LPIPS	FG-SSIM	CIEDE2000
GT-256 reference bake	22.38	0.0590	0.7683	4.44
Native residual†	11.13	0.1372	0.3376	26.47
No Adapter†	5.92	0.1600	0.3829	40.92
Constant 1.0 (capped)	11.93	0.1354	0.3504	24.18
Fixed Low	12.48	0.1317	0.3835	22.71
Fixed High	12.29	0.1305	0.4587	23.27
Fixed Mean	12.62	0.1303	0.4193	22.32
C3 / Middle-high	12.48	0.1302	0.4059	22.16
Late-high	12.92	0.1277	0.4259	21.65
Generic Linear	11.60	0.1276	0.4458	24.71

object-bootstrap CI [2.3%,5.3%]). This ratio of paired means is not a theoretical ceiling, a new primary test or evidence of acceptable material fidelity. Leaving out each object in turn keeps the mean Late–Mean difference between −0.00294 and −0.00208, but all substantial reference-appearance discrepancies remain. Supplementary Section S5 supplies the paired values and calculation.

C3–Fixed High has mean PSNR +0.19 dB (95% CI [−0.65, +0.87]) and LPIPS −0.00036 ([−0.00396, +0.00336]); neither separates under the four-comparison secondary Holm family (both adjusted $p = 1$). Its SSIM is lower by 0.05274 ([−0.07459, −0.03170]; Holm $p = 0.00781$). Thus the old five-object C3 PSNR advantage does not repeat stably on this new pilot. We retain C3 as the historical candidate and do not rename Late-high as C3.

Across the seven conditions, mean within-object Spearman correlations between generated-view and baked unseen-view rankings are 0.846 for PSNR, 0.804 for LPIPS, 0.746 for SSIM and 0.907 for Edge-SSIM. Broad rankings are positively associated, while the winning condition changes on 7/10 objects for LPIPS (1/10 for PSNR). These are different statements: we do not claim a general reversal of rankings. Full-object Figure 4 also shows persistent appearance errors, including the Tire's brown output against a gray reference. A lower LPIPS does not establish faithful material reproduction.



Fig. 4. Actual source and unlit baked outputs at the same unseen camera, with whole-object framing and no color or detail editing. These are the first two input-order objects with verified publication licenses, not outcome-selected examples. All seven conditions plus source and GT-256 bake for seven rights-cleared objects appear in the complete bake displays in Supplementary Section S5; all ten objects enter the numerical analysis. LG Remote: mohd.akmal.mahyudin, CC Attribution; Tire: Koto3D Stephane Vogley, CC Attribution (full source links in Supplementary Section S5). The Tire illustrates a retained appearance failure.

Separate twenty-object GSO evaluation.. The GSO/MV-Adapter study completes twelve conditions by three seeds on twenty fixed inputs, plus twenty matched GT self-bakes: 740 unique baked meshes and 8,140 unseen-view records. The objects, source meshes, projection cameras, texture resolution and eleven unseen cameras are shared across controls; unseen cameras do not contribute to baking.

The original attempt stopped after fifteen objects because a discontinuous UV-ramp diagnostic failed on a Hair Dryer. Subsequent GT-only tests identify finite-resolution mixing across UV jumps and validate continuous nonlinear surface-color fields, an independent multi-island analytic fixture, and a deliberate image-flip negative control. The resulting calibration addendum is post hoc. It keeps projection, masks, cameras, material semantics, resolutions and the fixed twenty objects unchanged; all 375 original GLB hashes and metric records reproduce exactly. The original failure and remaining UV/coverage limitations are retained in Supplementary Section S4.

Late-high minus Fixed Mean unseen-view LPIPS is +0.000084 (pointwise interval $[-0.000103, +0.000286]$; original comparison-family Holm $p = 0.788$). This study does not detect the ten-object pilot's benefit. It changes both input domain and conditioning/projection protocol, so it is neither a direct same-protocol replication nor evidence assigning the difference to backbone alone. Table 5 presents both application comparisons.

Across all twelve GSO conditions, mean within-object image/bake LPIPS-rank correlation is 0.601 (object interval $[0.440, 0.733]$). A post hoc sensitivity restricted to the seven original scale/time controls gives 0.396 ($[0.175, 0.582]$), with the minimum-mean LPIPS condition changing on 11/20 objects. These are positively associated rankings with changing point-estimate winners, not general rank reversal or statistically resolved winner changes. The GT self-bake has mean LPIPS 0.0602 and PSNR 19.36 dB; average complete-bilinear-footprint coverage is about 78.7%. All foreground remains scored and GT errors are not subtracted. These losses contextualize the small method effects without defining a theoretical quality bound.

Native controls and protocol sensitivity.. Subsequently added Native residual and No Adapter outputs on the same ten GeoTex objects are exploratory. They retain matched geometry and GT coverage, and the repeated Constant 1.0 (capped) outputs are bitwise identical. Their apparent PSNR/LPIPS gains for the schedules do not separate under the fifteen-test exploratory family. The complete data are in Supplementary Section S5.

A supporting fifty-object comparison uses a newly trained frozen GeoTex checkpoint with 512-pixel supervision, actual positive/negative reference conditions and CFG 2. Late-high minus Fixed Mean has nonreference PSNR -1.184 dB (interval $[-1.587, -0.849]$) and LPIPS +0.00266 ($[-0.00003, +0.00640]$; adjusted $p = 0.080$). C3 has no corrected advantage over Fixed Mean or Fixed High. This previously studied cohort and changed checkpoint/protocol supply a sensitivity boundary, not an independent backbone confirmation. Training and full-v29 strong-baseline results remain in Supplementary Section S8.

Table 5. Late-high minus Fixed Mean on unseen-view foreground LPIPS. Lower is favorable. All views/seeds are nested within object. The ten-object specified primary uses its original exact sign-flip test; GSO uses the original Holm16 comparison family. Pointwise intervals are not simultaneous. The two cohorts and protocols are not pooled.

Study	Objects	Fixed Mean	Late-high	Paired difference [95% CI]	Test
GeoTex compatibility pilot	10	0.1303	0.1277	−0.002652 [−0.004144, −0.001367]	exact $p = 0.00195$
GSO revised-calibration study	20	0.10089	0.10098	+0.000084 [−0.000103, +0.000286]	Holm $p = 0.788$

4.6. Transfer to other conditioning interfaces

MV-Adapter and MVDiffusion supply different conditioning operators and scale semantics from GeoTex. We evaluate each within its native target/camera contract and do not pool their scores or interpret identical scale labels as common residual doses.

The historical 76-object MV-Adapter layer/time panel has mixed endpoints and unadjusted temporal intervals. In the 75-object MVDiffusion panel, the layer-wise L–LLH condition is worse than global fixed low in PSNR, structural agreement and reference-color error; LPIPS does not separate. The intervention scales correspondence blocks rather than the same GeoTex residual operator. These counter-results, including all contrasts and identity-check scope, remain in Supplementary Section S3. They limit transfer claims; low absolute quality is not grounds for excluding them.

New native MV-Adapter temporal comparison.. We froze fifty Fresh C inputs not previously inspected through MV-Adapter, excluding all recorded earlier native and profile-panel objects. The input-only ordered selection used availability and native material/geometry compatibility. The native mesh normalization, six orthographic cameras and nvdiffrast rasterizer also generate evaluation targets; an independent source-UV sampling check validates this target path. Three textured new objects pass official/deployed and constant-default-path identity checks: 18 RGB images, conditioning features and injected residuals are bitwise identical. The old seven-object quality archive fails camera calibration and is superseded, not pooled.

The seven conditions share object inputs, seed, geometry and evaluation code. With 50 steps, low/high are 0.75/1.0; each stage-high schedule has 16 high steps, Fixed Mean is 0.83, and Generic Linear is 0.75–0.91. These five mean-matched controls each sum to 41.5 per scaled feature, within this interface. The fifty-object matrix is complete (350 rows), including thirteen texture-absent sources with legal constant-white base-color targets and a documented loader compatibility path. The other 37 textured objects are a descriptive subset, not a replacement primary cohort. Targets are unlit native-camera base color, not Fresh C's differently normalized reference renders or certified physical albedo.

The internally predeclared primary, Middle-high minus Fixed Mean FG-PSNR, is +0.029 dB (95% object-bootstrap CI [−0.019, +0.081]; paired Monte Carlo sign-flip $p = 0.281$; 29/50 favorable). This does not support a stable middle-window gain or establish equivalence. The remaining 104 pair–metric tests use one Holm family. Middle-high has lower Edge-SSIM than Fixed Mean (adjusted $p = 0.00104$); some other placement contrasts separate, including Late-high versus Early-high PSNR. Such endpoint-dependent differences do not demonstrate transferable C3 optimality. Complete contrasts, correction rules and the textured subset are in Supplementary Section S3. This is a native-interface validation of the scheduling question, not architecture-matched replication of GeoTex's capped residual intervention.

Table 6. Native-camera MV-Adapter comparison, fifty input-selected objects. Views are averaged within object. Scale values and endpoints are specific to this interface, and are not numerical doses shared with GeoTex.

Condition	FG-PSNR	FG-LPIPS	FG-SSIM	Edge-SSIM
Native / Fixed High	12.60	0.1280	0.5041	0.3526
Fixed Low	12.73	0.1270	0.5066	0.3475
Fixed Mean	12.68	0.1279	0.5068	0.3509
Early-high	12.60	0.1296	0.5041	0.3503
Middle-high	12.71	0.1262	0.5061	0.3470
Late-high	12.74	0.1260	0.5075	0.3514
Generic Linear	12.72	0.1263	0.5072	0.3501

External-input native schedule comparison.. The source-ordered GSO100 convenience cohort uses the same native MV-Adapter control path and three shared seeds. Native/Fixed High, Fixed Low, Fixed Mean, Early-high, Middle-high, Late-high and Generic Linear include the four inherited profiles frozen before their outputs, after earlier baseline exposure. The full dataset contains 3,600 generated cells and 21,600 view records, including three archived projection corrections and two separately frozen APG baselines [38]. The scheduling extension is external sensitivity, not pristine independent confirmation or parameter search.

The specified all-six-view Middle-high minus Fixed Mean primary is −0.014 dB (interval [−0.031, +0.004]; paired $p = 0.122$). A secondary Late-high–Fixed Mean contrast is +0.045 dB ([+0.024, +0.065]; Holm83 $p = 0.00335$), with SSIM +0.00134 (adjusted $p = 0.000415$). Its five-nonreference-view LPIPS difference is −0.000086 ([−0.000551, +0.000413]; separate Holm21 $p = 1$). Thus a small detected image effect does not establish a perceptual or baked-appearance benefit. Table 7 and Figure 5 put these named comparisons in context.

All fifteen specified GSO projection-minus-baseline nonreference-LPIPS comparisons favor the scale/APG baselines after their original separate Holm9/Holm6 corrections. The three projection candidates also lose to native on baked LPIPS in the retained twenty-object family. These valid negative auxiliary results are reported completely in Supplementary Sections S4 and S8; they are not proposed as additional methods.

4.7. Historical Preference and Training Evidence

The original blinded 3AFC study on thirty objects and twenty-four participants reports C3's overall-quality

Table 7. Selected within-study paired effects. P: the specified primary, unadjusted within its study; S: secondary with its original family adjustment (GSO all-six Holm83, nonreference LPIPS Holm21, reference-protocol secondary Holm18). Intervals are pointwise. Native50 and GSO100 use different inputs/seed policies; the reference-protocol study also changes weights and sampling and is not a separate backbone.

Study	Contrast	Endpoint	Mean	95% CI	Class	p
Native MV-Adapter50	Middle–Mean	FG-PSNR	+0.029	[-0.019,+0.081]	P	0.281
GSO100	Middle–Mean	FG-PSNR	-0.014	[-0.031,+0.004]	P	0.122
GSO100	Late–Mean	FG-PSNR	+0.045	[+0.024,+0.065]	S	0.003
GSO100	Late–Mean	Nonref LPIPS	-0.000086	[-0.000551,+0.000413]	S	1.000
Reference-protocol50	Late–Mean	Nonref PSNR	-1.184	[-1.587,-0.849]	S	< 0.0001

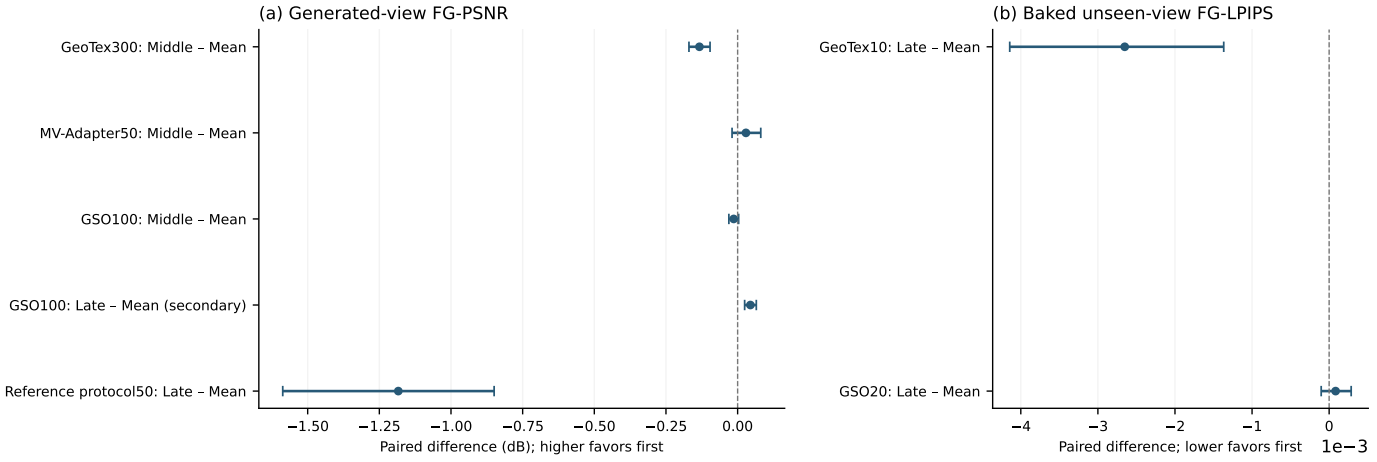


Fig. 5. Paired allocation effects from completed object-level records. Error bars are pointwise 95% object intervals. Panel (a) uses six-view PSNR except the reference-protocol row, which excludes its reference view; panel (b) uses eleven unseen baked views. The GSO Late–Mean PSNR row is secondary and is explicitly distinguished from the specified middle-window questions. Different studies are shown separately, without pooling or cross-pipeline absolute-score ranking. The small GeoTex bake gain and GSO non-detection both remain visible.

share as 58.1% (two-level participant/object-bootstrap interval [50.1, 65.8]%). This supports overall preference for the original stimuli, not texture fidelity, the retrained checkpoint or unseen baked appearance. Counts and endpoint-specific results are retained in Supplementary Section S7.

FAC re-examination.. Previously reported FAC configurations all reduced FG-PSNR relative to TCAS in their stated controlled re-examination. The original training pool was reported as 1,706 objects, while a recovered configuration refers to 1,200; seed, checkpoint and sampled-run records are incomplete. The results and reproducible inference/statistical components are separated from this remaining training-replay gap in Supplementary Section S7. Later new geometry training does not restore historical FAC identity.

5. Discussion and Limitations

What the controls establish.. The Fresh C300 gain over fixed low supports the recorded intervention under its frozen protocol. It changes both budget and timing. The equal-applied-budget matrix then measures a timing-allocation total effect and changes the preferred image-space operating point. It does not identify a state-independent residual dose or an optimal middle window. Competitive fixed and linear controls are necessary before assigning practical value to a selected profile.

Application value and operating points.. In the GeoTex image protocol, Fixed High and Generic Linear have stronger endpoint means than C3; equal-budget late-high trades higher PSNR and lower LPIPS against lower FG-SSIM than fixed mean. The ten-object bake's approximately 2% LPIPS reduction closes only 3.7% of the gap to the matched GT reference bake, while visible appearance errors remain. The separate GSO20 study does not detect that allocation benefit. Its post hoc calibration, GT self-bake loss and incomplete coverage are disclosed; a small image effect alone does not justify replacing an existing deployment setting.

Generated-view metrics remain informative: overall baked/image method rankings are positively associated. However, preferred point estimates often change, and the association is weaker when the GSO analysis is restricted to closely competing scale/time settings. Winner changes do not prove significant separation between methods or a general reversal. Actual baked evaluation and full-object reference comparison remain necessary.

Interface and protocol boundaries.. Native MV-Adapter50 and GSO100 do not detect the primary middle-window advantage. The secondary GSO late-window PSNR change is statistically detectable but small, with no corresponding nonreference LPIPS benefit. The supporting reference-conditioned checkpoint shows a different Late–Mean direction. Historical MVDiffusion supplies a partial-operator boundary. Because weights, inputs, conditioning operators and sampling protocols

differ, these studies cannot isolate one backbone or domain as the cause of the varying effects.

Quality, selection and replication limits.. The bake studies favor compatible native-UV assets, and GSO uses a source-ordered convenience sample. They do not establish representative asset quality, physical albedo, PBR fidelity or a validated seam score. Views and seeds are nested within object, and seed repetitions do not estimate training-checkpoint variation. Separate test families and pointwise intervals retain their stated roles; non-detection is not equivalence.

Fresh C uses retrained weights and a controlled sampler rather than the authenticated historical main adapter or complete default pipeline. The old +0.96 dB contrast is not reproduced, and historical FAC training records remain incomplete. The original preference study concerns its own stimuli. These limitations constrain deployment and replication claims, while the matched within-protocol comparisons remain valid.

6. Conclusion

Geometry-control allocation changes generated-view agreement, but its effects depend on the tested interface, protocol and endpoint. The matched 300-object study shows that C3's gain over fixed low combines increased strength with timing; equal-budget fixed mean and late-high outperform the historical middle window. Native MV-Adapter studies do not detect the primary middle-window advantage, while some secondary image effects are small but measurable. Actual baking yields a limited late-high benefit on ten compatible GeoTex objects and no detected LPIPS benefit on a separate twenty-object GSO cohort. Method rankings are broadly associated across image and baked evaluation, yet their preferred operating points often differ. Competitive controls and application-level evaluation are therefore necessary when choosing a setting within a deployed pipeline. The results do not identify a universally preferred schedule or establish faithful material reproduction.

Data availability

Object-level revision metrics, frozen protocols/configuration records, verified inference and analysis scripts, and the LaTeX source required to rebuild the manuscript and Supplement are included in the author-review package. The standalone reproduction archive supports statistic recomputation without model weights; fresh inference additionally requires the recorded weights, runtime and licensed source assets. The complete raw generated-image archives, trained checkpoints, and participant-level responses are not included. Source assets must be acquired under their recorded licenses; the review evidence includes attributed displays and a small licensed GSO example export. Additional analysis records may be requested from the corresponding author, subject to licensing, privacy, and archive constraints.

Declaration of generative AI and AI-assisted technologies in manuscript preparation

OpenAI Codex assisted with revision drafting, implementation and reproducible statistical and figure code. OpenAI Images supplied conceptual artwork for Figure 1; the service did not expose a model version in the preserved generation record. Actual input/output thumbnails and experimental displays come from the recorded research runs and were not retouched with an image-generation tool. Numerical tables are computed from delivered object-level records. Final scientific review, accuracy and responsibility remain with the authors.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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